Machine Learning Foundations

Week 2 Session

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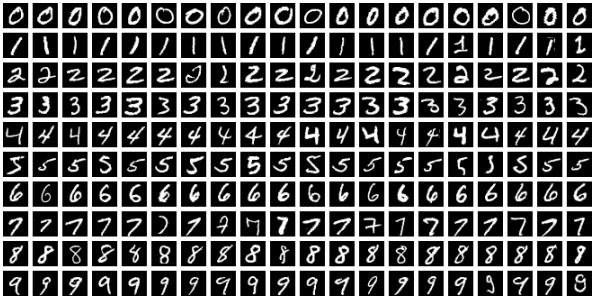
Start with a picture, not a formula



What would a computer need to learn before it could recognize this as a cat?

Take a moment. What would you look at first?

A model does not see the word “cat”



A computer receives numbers that describe an image.

People may call the image a handwritten 8. The model must use its numerical input to produce a useful answer.

Input: pixel values about the digit

Output: a guess

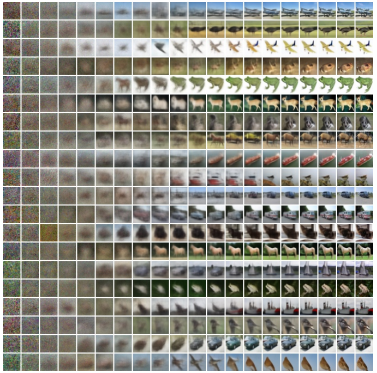
Different tasks, same learning story

Next word

"The cat sat on the ..."

The model uses the words so far to make a guess about what comes next.

New image



A generated image can gradually emerge from noise.

Recognize a digit

Which patterns in the pixels suggest an 8 rather than a 3?

Different applications, but each model learns how numerical inputs relate to useful outputs.

Learning means adjusting many numerical choices

A neural network contains adjustable numbers called parameters.

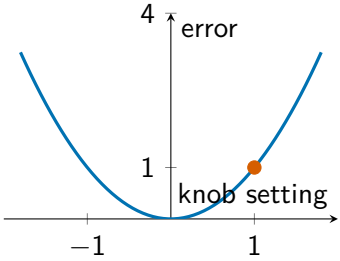
- The parameters affect the model's answer.
- The answer can be compared with the answer we wanted.
- Training changes parameters so future answers improve.

The central question is simple to say: if I change a parameter a little, how will the error change?

Why calculus? A one knob thought experiment

Imagine that a model has one adjustable knob. Each setting gives a different training error.

We want to know which nearby setting makes the error smaller. Calculus gives us a language for local change.

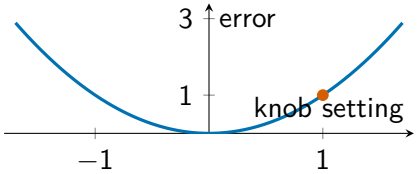


Poll: a small move in the right direction

Poll Question

The curve shows error as one model knob changes. We are at knob setting 1. Which small move should lower the error?

A) Move left B) Move right C) Either direction D) There is not enough information



The same question appears in backpropagation

A neural network can have millions of parameters, but training still asks how each small parameter change affects the error.

- For one parameter, we ask for a slope.
- For many parameters, we collect those slopes into a gradient.
- The chain rule lets us trace how a change at an early stage affects the final error.

Today we will build the mathematical language behind those ideas, one piece at a time.

From numbers to functions

First we need a clear way to say what kinds of inputs and outputs we are talking about.

Numbers, nonnegative numbers, and tuples

- $\mathbb{R}$ is the set of real numbers: -2, 0, $\frac{1}{2}$, $\sqrt{2}$, and so on.
- $\mathbb{R}_+ = [0, \infty)$ is the set of nonnegative real numbers. Here zero is included.
- $\mathbb{R}^n$ is the set of ordered lists of n real numbers.

$$(2, -1) \in \mathbb{R}^2, \qquad (0.2, 0.7, 0.1) \in \mathbb{R}^3$$

An image with many pixels can be represented by a long vector of numbers.

Sets give us a language for collections

A set is a collection of objects. The objects may be numbers, vectors, or other items.

$A = \{1, 2, 3\}$ and $B = \{2, 3, 4\}$.

$2 \in A$ means 2 is in A. $5 \notin A$ means 5 is not in A.

$$A \cup B = \{1, 2, 3, 4\}$$

Union collects elements in either set.

$$A \cap B = \{2, 3\}$$

Intersection collects the elements they share.

In machine learning, a set often describes all allowed inputs or all possible outputs.

What makes a vector space?

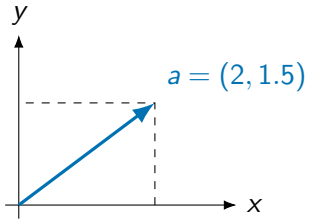
A vector space is a collection where we can add vectors and scale them, and stay inside the collection.

- If $a, b \in V$, then $a + b \in V$.
- If $a \in V$ and $\lambda \in \mathbb{R}$, then $\lambda a \in V$.
- There is a zero vector, and addition and scaling obey familiar rules.

$$\lambda a + \mu b \in V$$

This is a linear combination of a and b. For example, $\mathbb{R}^2$ and $\mathbb{R}^3$ are vector spaces.

A vector can be a direction, a location, or a list of features



A vector in $\mathbb{R}^n$ is an ordered list of n numbers.

A point, an image, or a collection of measurements can all be encoded this way.

Dot product: how aligned are two vectors?

For $a, b \in \mathbb{R}^n$, the dot product is

$$a \cdot b = \sum_{i=1}^{n} a_i b_i.$$

Example:
$(1, 2) \cdot (3, 4) = 1 \cdot 3 + 2 \cdot 4 = 11.$

If $a \cdot b = 0$ and neither vector is zero, the vectors are perpendicular.

The dot product will later help us measure change in a chosen direction.

Length measures the size of a vector

The Euclidean length, or norm, of $a = (a_1, \ldots, a_n)$ is

$$\|a\| = \sqrt{a \cdot a} = \sqrt{a_1^2 + \cdots + a_n^2}.$$

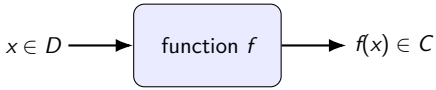
$$\|(3, 4)\| = \sqrt{3^2 + 4^2} = 5.$$

This is the familiar Pythagorean distance, extended to any number of coordinates.

A function is a machine with a rule

A function takes an allowed input and assigns it exactly one output.

Domain: allowed inputs
Range space: where outputs live
Rule: how the output is assigned



The input dimension and output dimension

We write

$$f : \mathbb{R}^d \longrightarrow \mathbb{R}^m.$$

- d tells us how many numbers describe one input.
- m tells us how many numbers the function returns.

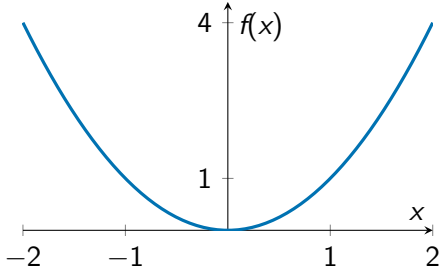
$$f : \mathbb{R}^2 \to \mathbb{R}, \qquad f(x, y) = x^2 + y^2$$

Two input coordinates, one output number. This is a scalar valued function of two variables.

One input: a graph is a curve

For $f : \mathbb{R} \to \mathbb{R}$, we can plot input x against output $f(x)$.

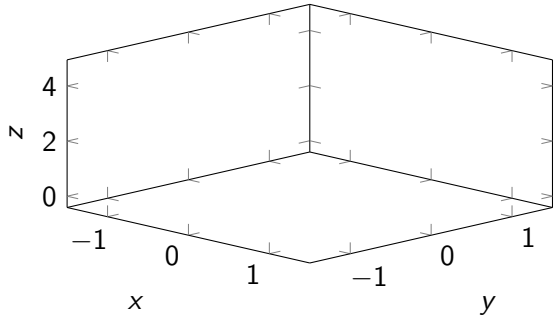
Example: $f(x) = x^2$.



Two inputs: the graph becomes a surface

For $f : \mathbb{R}^2 \to \mathbb{R}$, the graph uses two input axes and one output axis.

Here $z = f(x, y) = x^2 + y^2$.



Three inputs: already hard to draw

A function $f : \mathbb{R}^3 \to \mathbb{R}$ needs three input coordinates and one output value.

We cannot place all four coordinates on an ordinary flat page at once.

But we can ask a simpler question: where is the output the same?

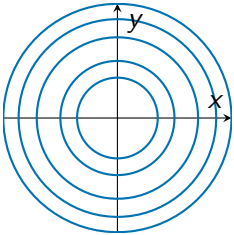
That question leads to contour plots.

A contour is a level set drawn on the input plane

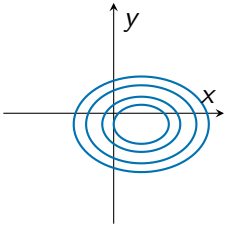
Fix an output value c. The contour at level c is the set of inputs satisfying

$$f(x, y) = c.$$

Each contour joins locations with equal output.



Read the contours as a map of height



Close contours mean the output changes quickly across the map.

Wide spacing means it changes more slowly. A contour plot is a compressed way to see a surface from above.

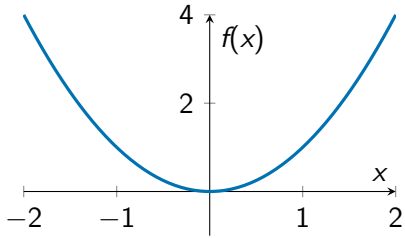
A first idea of continuity

We will begin with a picture in our heads, then translate it into limits.

Continuity: the no surprise idea

A continuous curve has no sudden jump, hole, or break.

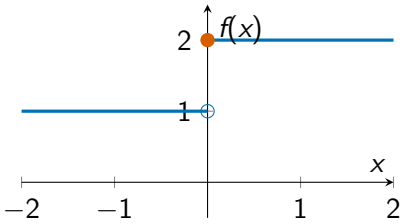
Informally, near a point, nearby inputs give nearby outputs.



A jump fails the no surprise test

As we approach zero from the left, the graph approaches one value. From the right, it approaches another.

The two sides do not meet.



Now say the same idea in limit language

A function f is continuous at a when all three statements hold:

$$\lim_{x \to a} f(x) \text{ exists}, \qquad f(a) \text{ is defined}, \qquad \lim_{x \to a} f(x) = f(a).$$

For a two sided limit to exist, the one sided limits must agree:

$$\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x).$$

In words: left and right must meet at the actual value of the function.

Example: where can a piecewise rule break?

Consider

$$f(x) = \begin{cases} x + 1, & x < 0, \\ 2x + 1, & x \geq 0. \end{cases}$$

Away from zero, each rule is a line, so there is no break inside either piece.

$$\lim_{x \to 0^-} f(x) = 1, \qquad \lim_{x \to 0^+} f(x) = 1, \qquad f(0) = 1.$$

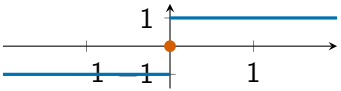
The pieces meet, so f is continuous at zero.

Poll: continuity of the sign function

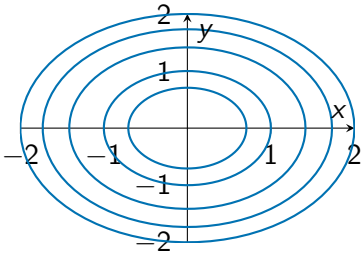
Poll Question

Define $\text{sgn}(x) = -1$ for $x < 0$, 0 for $x = 0$, and 1 for $x > 0$. Is sgn continuous at zero?

A) Yes, because its value is defined at zero
B) Yes, because the graph has only three possible values
C) No, because the left and right limits disagree
D) No, because it is not defined at zero



A contour picture can look broken without the function jumping



Poll: is the underlying function continuous?

The curves are separate contour levels. They do not mean the surface has gaps between them. A smooth bowl has continuously many heights.

A contour map samples levels, much like elevation lines on a map.

Poll: what does a contour gap mean?

Poll Question

The plotted circles are several level sets of $f(x, y) = x^2 + y^2$. Does the empty space between circles imply a discontinuity?

A) Yes, the function is undefined between contours
B) Yes, each circle is a separate piece of the surface
C) No, the contour lines show selected output levels of a continuous surface
D) There is no function because the picture is two dimensional



PYQ 1: choose the value that joins two pieces

Selected PYQ

Find k so that the piecewise function is continuous at $x = 2$:

$$f(x) = \begin{cases} 3x^2 - kx + 5, & 0 \le x < 2, \\ 1 - 3x, & 2 \le x \le 3. \end{cases}$$

Enter the value of k that makes f continuous.

Pause and set the left hand limit equal to the value from the right branch.

PYQ 2: the value at zero cannot repair a jump

Selected PYQ

Let f be defined by

$$f(x) = \begin{cases} \dfrac{x - |x|}{x}, & x \neq 0, \\ 2, & x = 0. \end{cases}$$

Is f continuous? Enter 1 for Yes and 0 for No.

PYQ 3: can an oscillating function settle down?

Selected PYQ

Does $\lim_{x \to 0} \sin(1/x)$ exist? Enter 1 for Yes and 0 for No.

Think about what happens when $1/x$ lands repeatedly at the top and bottom of the sine wave.

PYQ 4: continuity is required for differentiability

Selected PYQ

Let $f(x) = |x|/x^2$ for $x \neq 0$ and $f(0) = 0$. Which statements are true?

A) f is differentiable on $(-\infty, 0)$ and $(0, \infty)$
B) f is continuous at zero
C) f is differentiable at zero
D) f is discontinuous at zero

The derivative is a local slope

We can make a curved graph feel less mysterious by zooming in near one point.

Differentiability asks for one stable tangent slope

At $x = a$, compare the function value at $a + h$ with its value at a.

$$\text{average rate of change} = \frac{f(a+h) - f(a)}{h}.$$

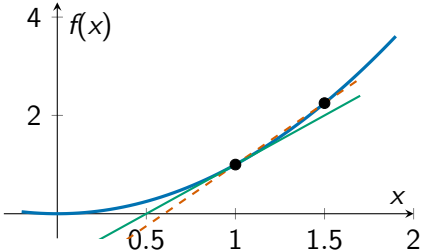
If this settles to one number as h shrinks to zero, that number is the derivative:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

The derivative is the instantaneous rate of change, or the slope of the tangent line.

A secant line becomes the tangent line

The secant uses two points.
Move the second point toward the first, and its slope approaches the tangent slope.



The dashed red line is a secant. As its second point moves toward $(1, 1)$, it approaches the green tangent of slope $f'(1) = 2$.

PYQ 5: find a quadratic curve from its tangent

Selected PYQ

The curve $y = ax^2 + bx + c$ passes through $(2, -1)$ and $(0, 0)$. It is tangent to $y = -x$ at the origin. Find $|ab|$.

Two friendly derivative examples

For $f(x) = x^2$:

$$f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$
$$= \lim_{h \to 0} (2x + h) = 2x.$$

For $g(x) = 3x + 1$, every secant slope is 3, so $g'(x) = 3$.

Curves can have a different local slope at every point. A line keeps the same slope everywhere.

The derivative gives a nearby straight line

Near $x = a$, a differentiable function behaves approximately like its tangent:

$$f(a + h) \approx f(a) + f'(a)h.$$

Equivalently, for an input x near a,

$$f(x) \approx f(a) + f'(a)(x - a).$$

This is the first order linear approximation. It is useful because a line is easy to calculate with.

Why the linear approximation is reasonable

By the definition of derivative,

$$\frac{f(a+h)-f(a)}{h} \approx f'(a)$$

when h is small.

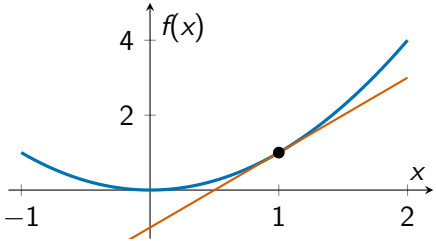
$$f(a+h)-f(a) \approx f'(a)h \quad \Longrightarrow \quad f(a+h) \approx f(a)+f'(a)h.$$

We have not guessed the tangent formula. It follows directly from the limit definition.

A picture of local approximation

Far from the chosen point, a tangent line may be a poor stand in.

Very close to the point, the curve and its tangent nearly overlap.



Higher order approximation remembers curvature

A tangent uses the value and slope. A quadratic approximation also uses curvature:

$$f(a+h) \approx f(a) + f'(a)h + \frac{f''(a)}{2}h^2.$$

More generally, Taylor's formula continues with cubic and higher powers when more accuracy is needed.

Today we will keep terms through h^2. That is enough to explain curvature and several useful rules.

Multiplying two local approximations gives the product rule

Near a, write

$$f(a+h) \approx f(a) + f'(a)h, \qquad g(a+h) \approx g(a) + g'(a)h.$$

Multiply and ignore terms in h^2 and above:

$$f(a+h)g(a+h) \approx f(a)g(a) + h\big(f'(a)g(a) + f(a)g'(a)\big).$$

Therefore $(fg)'(a) = f'(a)g(a) + f(a)g'(a)$.

Composing two local approximations gives the chain rule

Suppose $z = g(x)$ and $y = f(z)$. A small change h in x gives

$$\Delta z \approx g'(x)h.$$

The resulting output change is

$$\Delta y \approx f'(z)\Delta z \approx f'(g(x))g'(x)h.$$

$$\boxed{\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)}$$

The inner change happens first, then the outer function responds to it.

Poll: estimate 1.1^7 without a calculator

Poll Question

Use the first order approximation of $f(x) = x^7$ around $x = 1$. Which estimate is closest to 1.1^7?

A) 1.07 B) 1.7 C) 7.1 D) 11

First find $f(1)$ and $f'(1)$. Then use the small change $h = 0.1$.

PYQ 6: use a tangent to estimate a fourth root

Selected PYQ

Use a linear approximation to estimate $\sqrt[4]{16.1}$, expanding around 16.

A) 2.1
B) 2.003125
C) 2.031
D) 1.9968

Local peaks, valleys, and a saddle

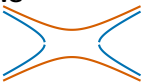
Minimum

A valley floor 

Maximum

A hilltop 

Saddle

A horse saddle 

Critical points are places worth inspecting

For a differentiable function of one variable, a local peak or valley often occurs where

$$f'(x) = 0.$$

In several variables, the corresponding condition is that every first partial derivative is zero.

$$\nabla f(x, y) = 0.$$

A critical point could be a minimum, maximum, or saddle. Later, the Hessian and eigenvalues will help distinguish them.

From slope to gradient

The one input story still works. Now we can move in many possible directions.

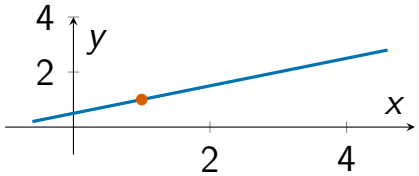
A line through a point, moving along a vector

Start at point p and move in direction v:

$$r(t) = p + tv.$$

For example, $p = (1, 1)$ and $v = (2, 1)$ give

$$r(t) = (1, 1) + t(2, 1).$$

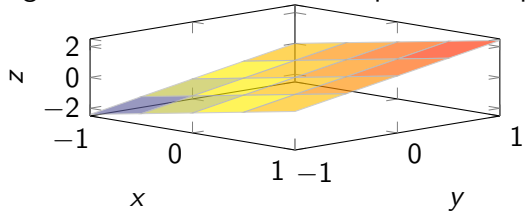


A plane through a point, spanned by two vectors

Start at p and move along either of two directions u and v:

$$r(s,t) = p + su + tv.$$

One parameter moves along one direction. The second parameter opens the line into a plane.



A hyperplane is a flat surface in higher dimensions

In $\mathbb{R}^3$, a plane is described by

$$ax + by + cz = d.$$

A hyperplane is the same kind of flat object in $\mathbb{R}^n$.

$$a_1 x_1 + \cdots + a_n x_n = d.$$

We cannot draw all its coordinates, but the equation and vector notation carry over unchanged.

Partial derivatives: change one input at a time

Imagine standing on a landscape.

- Walk east while holding north position fixed. How fast does height change?
- Walk north while holding east position fixed. How fast does height change?

$$f_x = \text{rate of change in the } x \text{ direction}$$

$$f_y = \text{rate of change in the } y \text{ direction}$$

The x partial changes x and freezes y.

The y partial changes y and freezes x.

The limit definition of a partial derivative

For $f : \mathbb{R}^2 \to \mathbb{R}$,

$$f_x(a, b) = \lim_{h \to 0} \frac{f(a + h, b) - f(a, b)}{h},$$

$$f_y(a, b) = \lim_{h \to 0} \frac{f(a, b + h) - f(a, b)}{h}.$$

The x partial changes x and freezes y. The y partial changes y and freezes x.

Example: partial derivatives of a bowl

Let $f(x, y) = x^2 + y^2$.

$$f_x(x, y) = 2x, \qquad f_y(x, y) = 2y.$$

At $(1, 2)$,

$$f_x(1, 2) = 2, \qquad f_y(1, 2) = 4.$$

If we move only in x, the local rate is 2. If we move only in y, it is 4.

The gradient gathers all the partial derivatives

For a scalar valued function $f : \mathbb{R}^d \to \mathbb{R}$,

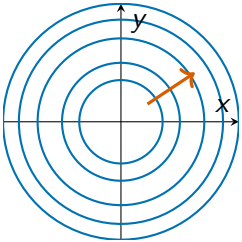
$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_d} \right).$$

It points toward the direction of steepest local increase.

$$f(x, y) = x^2 + y^2 \quad \Longrightarrow \quad \nabla f(x, y) = (2x, 2y).$$

The gradient is a vector of slopes, one for each input coordinate.

A gradient points uphill across contour lines



The gradient is perpendicular to the contour through the point.

It points toward increasing height. Its length tells how steep that uphill direction is locally.

Backpropagation speaks the language of gradients

A model's loss depends on many parameters:

$$\mathcal{L} = \mathcal{L}(\theta_1, \theta_2, \ldots, \theta_n).$$

The gradient

$$\nabla_\theta \mathcal{L} = \left(\frac{\partial \mathcal{L}}{\partial \theta_1}, \ldots, \frac{\partial \mathcal{L}}{\partial \theta_n} \right)$$

tells us how loss changes with each parameter. Backpropagation efficiently computes these derivatives using the chain rule.

Linear approximation in several variables

For a small input change $h = (h_1, \ldots, h_d)$,

$$f(x + h) \approx f(x) + \nabla f(x) \cdot h.$$

This is the multivariable version of

$$f(a + h) \approx f(a) + f'(a)h.$$

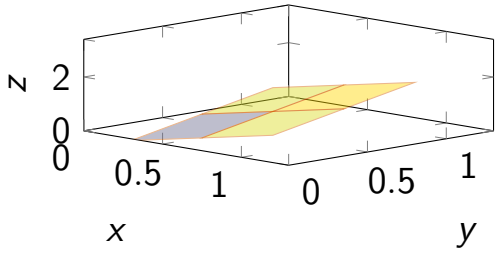
The dot product adds up the contributions from changes in every coordinate.

A tangent plane is the multivariable tangent line

For $z = f(x, y)$ near (a, b),

$$z \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b).$$

This equation describes a plane touching the surface at $(a, b, f(a, b))$.



Directional derivatives ask about a chosen walk

The gradient tells us how to move in the steepest uphill direction. What if we choose a direction u?

$$D_u f(x) = \lim_{h \to 0} \frac{f(x + hu) - f(x)}{h}.$$

If u is a unit vector, this is the rate of change per unit distance in direction u.

The directional derivative is a dot product

Use the local linear approximation:

$$f(x + hu) \approx f(x) + \nabla f(x) \cdot (hu).$$

Subtract $f(x)$ and divide by h:

$$D_u f(x) = \nabla f(x) \cdot u.$$

A dot product measures how much the gradient points along the direction we chose.

Cauchy Schwarz: projection cannot exceed length

The dot product $a \cdot b$ measures the part of a that points along b.

$$|a \cdot b| = \|a\| \, \|b\| \, |\cos\theta| \leq \|a\| \, \|b\|.$$

Why does this bound hold? The angle between two vectors has $|\cos\theta| \leq 1$. There is also an algebraic proof from the fact that a squared length cannot be negative.

An algebraic proof of Cauchy Schwarz

The squared length of $a - tb$ is nonnegative for every real t:

$$0 \leq \|a - tb\|^2 = \|a\|^2 - 2t(a \cdot b) + t^2\|b\|^2.$$

This quadratic cannot dip below zero. Its discriminant is therefore at most zero:

$$4(a \cdot b)^2 - 4\|a\|^2\|b\|^2 \leq 0.$$

$$\boxed{|a \cdot b| \leq \|a\| \, \|b\|}$$

Why Cauchy Schwarz identifies steepest ascent

For a unit direction u,

$$D_u f = \nabla f \cdot u \leq \|\nabla f\| \, \|u\| = \|\nabla f\|.$$

Equality occurs when u points in the same direction as ∇f.

So the gradient is not just an arbitrary vector of partial derivatives. Among unit directions, it gives the largest increase.

Minimum, maximum, or saddle in many variables

At an interior local minimum or maximum of a differentiable function, every directional derivative must be zero.

$$\nabla f(x^*) = 0.$$

But a saddle can also have zero gradient. For example, $f(x, y) = x^2 - y^2$ rises along one axis and falls along the other.

First derivatives locate candidates. Curvature, through the Hessian, helps classify them.

PYQ 7: a second order approximation

Selected PYQ

Find the second order approximation of $f(x, y) = e^{2y-x}$ near $(0, 0)$.

A) $1 - x + 2y - 2xy + y^2 + \frac{x^2}{2}$
B) $1 - 2xy + 2y^2 + \frac{x^2}{2}$
C) $2y^2 + \frac{x^2}{2}$
D) $1 - x + 2y - 2xy + 2y^2 + \frac{x^2}{2}$

PYQ 8: a tangent plane for ye^{xy}

Selected PYQ

Find the linear approximation of $f(x, y) = ye^{xy}$ in the neighbourhood of $(0, 1)$.

A) $x + y$
B) $x + 2y$
C) $2x + y$
D) $2x + 3y$

PYQ 9: use the gradient to estimate a nearby value

Selected PYQ

For $f(x, y) = \ln(x^2 + y^2)$, use the linear approximation at $(1, 2)$ to estimate $f(1.02, 2.01)$. Write the answer correctly to four decimal places.

The thread running through today's ideas

We began with a model making a guess. Then we built a language for inputs, outputs, and change.

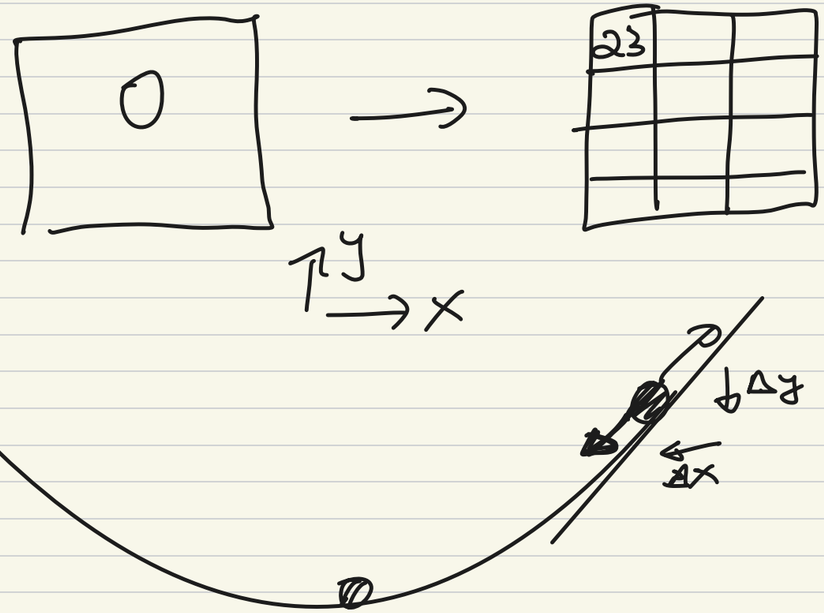
$$\text{local slope} \longrightarrow \text{derivative} \longrightarrow \text{gradient} \longrightarrow \text{backpropagation}$$

Next time these tools will help us reason about how learning algorithms improve a model.

Thank you!

See you next week.

Week-2 TA Session (Tuesday)



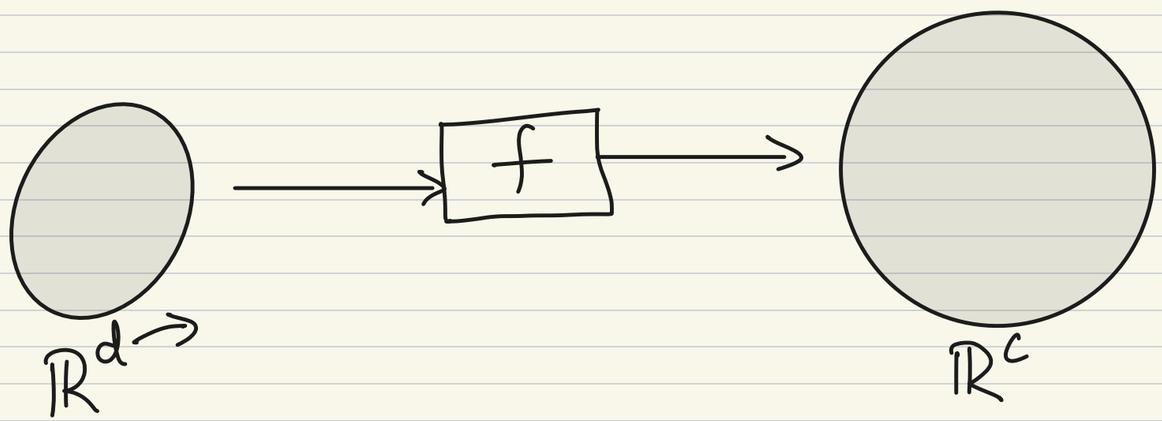
$\frac{dy}{dx}$ = Rate of change of y wrt x

Note

Please consider the presentation shared as the primary notes

Thanks ☺

* Function



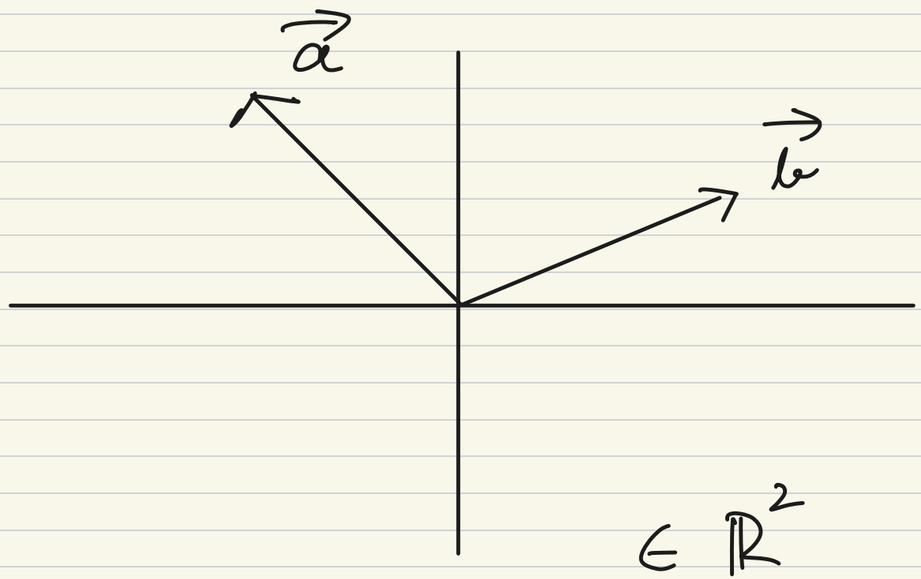
$\mathbb{R}^n$

Made with Goodnotes

$\mathbb{R}^n$: is the set of ordered list of n Real No.

$\mathbb{R}^3$: (a, b, c) — A Set is a collection of unique objects

* Vector Space

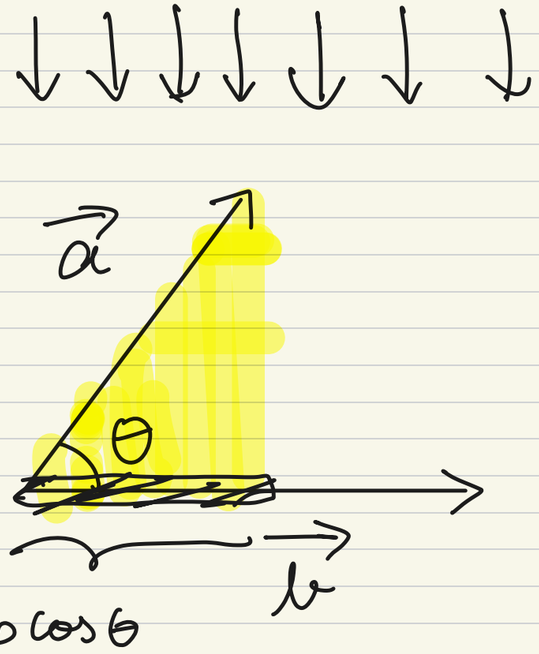


$\in \mathbb{R}^2$

- $a, b \in V$
  $\Downarrow$
  $a + b \in V$
  $\Downarrow$
  $\alpha a + \beta b \in V$
- $a \in V \Rightarrow \lambda a \in V$

Made with Goodnotes

* Dot product

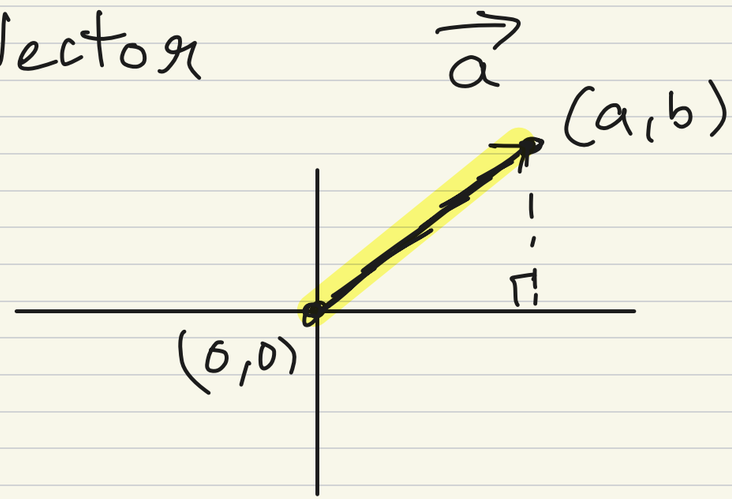


$$\vec{a} \cdot \vec{b} = \sum_{i=1}^{n} a_i b_i$$

Dot operator

$\mathbb{R}^2$: $a \cdot b \cdot \cos\theta$

* Norm of a Vector



$$\|a\| = \sqrt{a \cdot a}$$

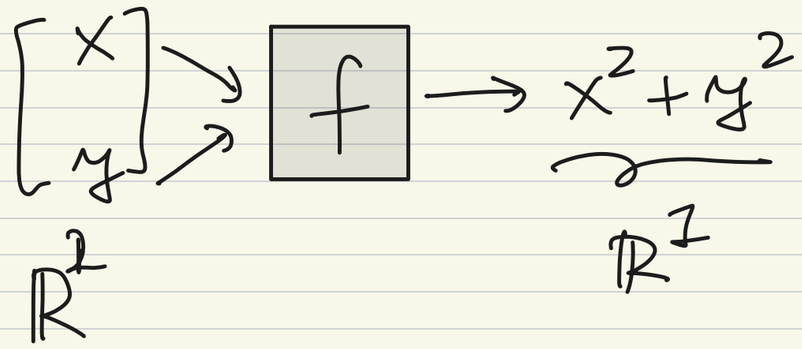
Norm of a vector

$$= \sqrt{a_1^2 + a_2^2 \cdots a_n^2}$$

Made with Goodnotes

Q] $f(x,y) = x^2 + y^2$

$f : \mathbb{R}^n \to \mathbb{R}^m$ $\quad (n, m) = ?$



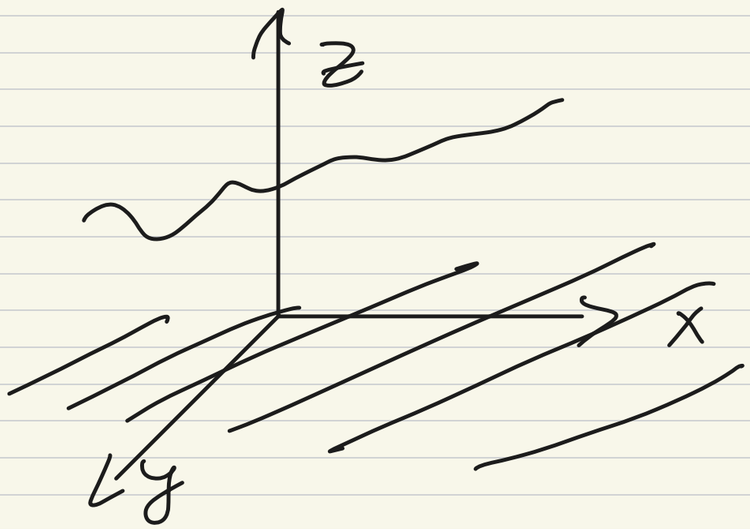
* Contour

$f(x,y) = c$

$f(x,y) = \underbrace{x^2 + y^2}$ $\quad (c)$

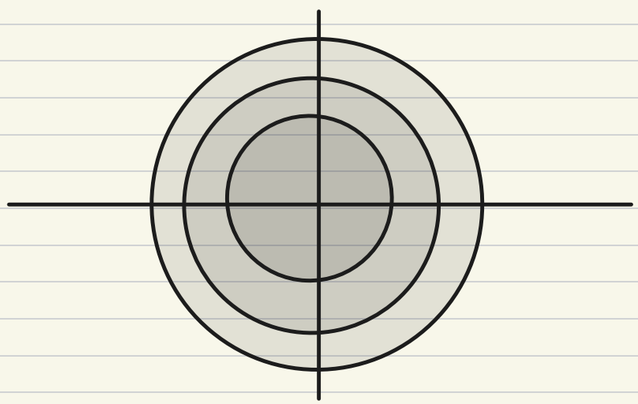
A level set at a const c

Made with Goodnotes



$\mathbb{R}^3 \to \mathbb{R}^2$

$x, y, z \to (x, y)$

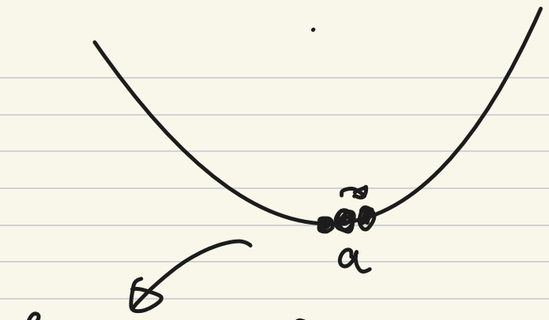


* Continuity

$\lim_{x \to a} f(x)$

$\lim_{x \to a}$

Made with Goodnotes

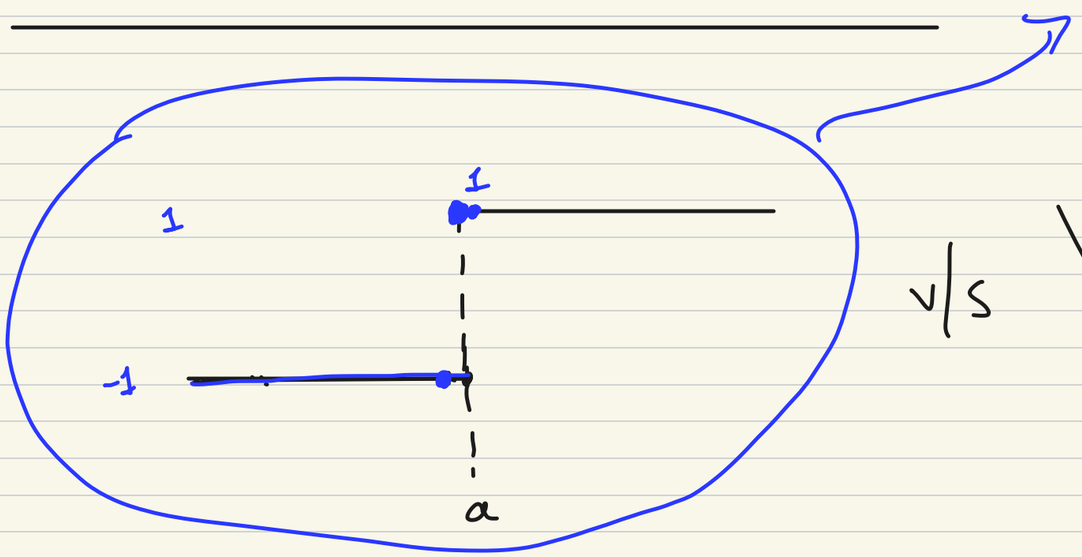


$\lim_{x \to a^-}$ $\lim_{x \to a^+}$

$a^- \to a - 0.0 - \cdots 0\ 01$

$a^+ \to a + 0.0 \cdots 01$

$a \to \boxed{f}$

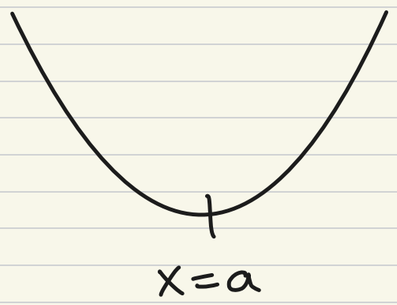


$f(a)$ is Not defined

✓ def 1 ✗

✓ def 2 LHL ≠ RHL

v/s



✓

v/s



✓

Q]

$$f(x) = \begin{cases} 3x^2 - kx + 5 & 0 \le x < 2 \\ 1 - 3x & 2 \le x \le 3 \end{cases}$$

given: $f \rightarrow$ Continious at $x = 2$, $K =$

Def^n - (2)

$$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x) = f(a)$$

$$x = 2 - 0.0 - \cdots 01 \simeq 2$$

LHL : $3x^2 - kx + 5$

Made with Goodnotes

$$LHL = 3(4) - K(2) + 5$$

$$\text{RHL:} \quad \lim_{x \to 2^+} f(x) = \lim_{x \to 2^+} (1 - 3x)$$

$$= 1 - 3(2)$$

Cont. $\Rightarrow$ $LHL = RHL$

$$3(4) - K(2) + 5 = 1 - 6$$

$$\boxed{K = 11}$$

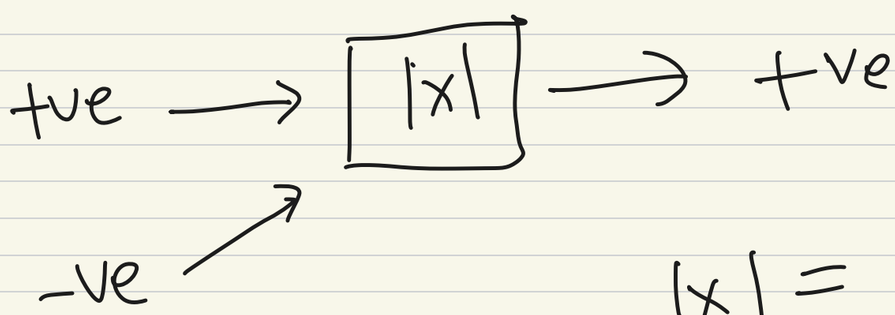
Made with Goodnotes

Q]

$$f(x) = \begin{cases} \dfrac{x - |x|}{x} & x \neq 0 \\ 2 & x = 0 \end{cases}$$

is $f(x)$ continious?

Yes / No ?



$$|x| = \begin{cases} x & \forall\, x \geq 0 \\ -x & \forall\, x < 0 \end{cases}$$

Made with Goodnotes

Q] $$f(x) = \begin{cases} \dfrac{x-|x|}{x} & x \neq 0 \\ 2 & x = 0 \end{cases}$$

is $f(x)$ continious?

Yes / No ?

Let us check limit at $\boxed{x=0}$

$\rightarrow$ LHL $\lim\limits_{x \to 0^-} f(x) \left(\dfrac{x-|x|}{x} \right)$

$$|x| = -x$$

$$\text{LHL} = \frac{x-(-x)}{x} = \frac{2x}{x} = 2$$

Made with Goodnotes

RHL $\lim_{x \to 0^+} f(x) = \frac{x - |x|}{x}$

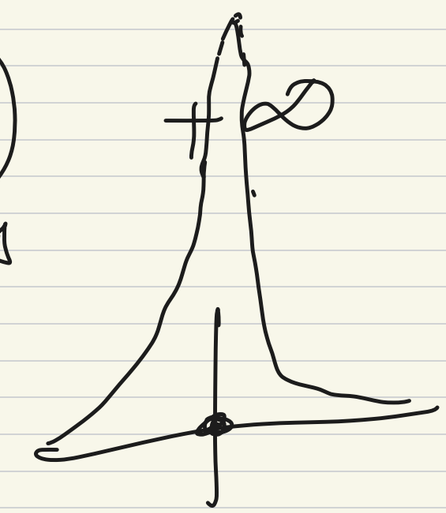
$= \frac{x - x}{x} = 0$

Q] $f(x) = \frac{|x|}{x^2}$ for $x \neq 0$ and $f(0) = 0$

check Continuity at $x = 0$

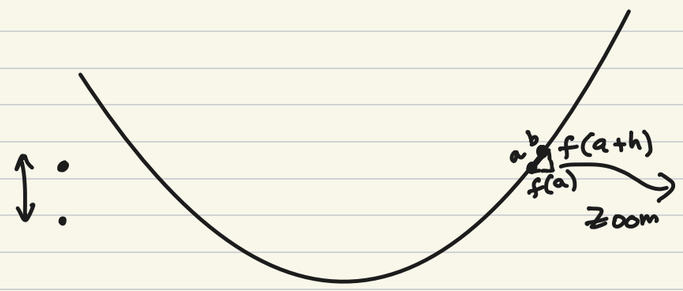
LHL $\lim_{x \to 0^-} \frac{-x}{x^2} = \frac{-1}{x} \to$ $-\infty$

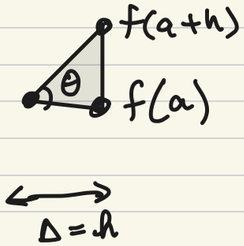
RHL $\lim_{x \to 0^+} \frac{x}{x^2} = \frac{1}{x} \to \infty$



Made with Goodnotes

* Derivatives





$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

$$\tan\theta = \frac{f(a+h)-f(a)}{h}$$

$$\rightarrow \frac{dy}{dx}$$

m

Q] $y = ax^2 + bx + c$ It is given that is passes through $(2,-1)$ & $(0,0)$. Tangent to $y = -x$ at $(0,0)$

$|ab|$

$y = -x$

(Recall: $y = \underset{-1}{m}x + c$)

$$\frac{dy}{dx} = 2ax + b$$

$$-1 = 2a(0) + b$$

$$b = -1$$

Made with Goodnotes

$y = ax^2 + bx + c \rightarrow (2, -1)$ & $(0,0)$

$0 = 0 + 0 + c$

$c = 0$

$-1 = a(4) + b(2)$

$-1 = 4a - 2$

$a = \frac{1}{4}$

$|ab| = \left|\frac{1}{4} \times -1\right| = \frac{1}{4}$

0.25

* Linear approximation

$$\left\{ f'(a) = \frac{f(a+h) - f(a)}{(h)} \right\} \quad -(1)$$

$$f'(a) \cdot h = f(a+h) - f(a) \quad -(2)$$

$$f(\widetilde{a+h}) \approx f(a) + f'(a) \cdot h \quad -(3)$$

$$\boxed{f(x) \approx f(a) + f'(a)(x-a)} + \underbrace{\quad}_{-(4)}$$

(x)

$$f(x) \underset{\sim}{\approx} f(a) + f'(a)(x-a)$$

Linear App

Made with Goodnotes

$$f(a+h) \simeq f(a) + f'(a)\,h + \frac{f''(a)\,h^2}{2!} + \cdots$$

"taylor series"

f^n

Made with Goodnotes

Q] $f(x) = x^7$ around $x = 1$

$$\frac{d}{dx} x^n = n x^{n-1}$$

$(1.1)^7$

$f(1) = 1^7 = 1$

$f'(x) = 7x^6 \Rightarrow (7)$ {at $x = 1$}

a) 1.07

b) 1.7

c) 7.1

d) 11

$1 + f'() \cdot (0.1) = 1 + 7(0.1)$

$=$

$(1.1)^7 = (1 + 0.1)^7$

Made with Goodnotes

Q] Linear approximation of $\sqrt[4]{16.1}$ around 16

$\{(16.1)^{1/4}\}$

$16.1 = \underbrace{16}_{a} + \underbrace{0.1}$

$$f(x) \approx f(a) + f'(a)(x-a)$$

$f(x) = x^{1/4}$

$f'(x) = \frac{1}{4}x^{-3/4} = \frac{1}{4x^{3/4}}$

$f'(16) = \frac{1}{4 \cdot 16^{3/4}} = \frac{1}{4(2^{4/4})^{x3}} = \frac{1}{4(8)} = \frac{1}{32}$

$16 = (2^4)^{3/4}$

Made with Goodnotes

$$\sqrt[4]{16.1} = 2 + (0.1)\left(\frac{1}{32}\right)$$

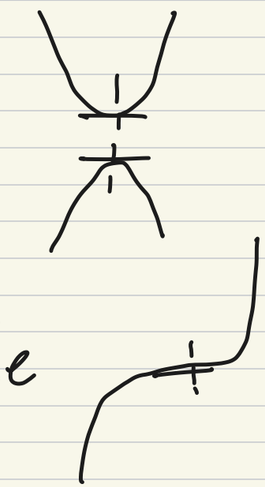
$$= 2.00$$

* Critical points

$f'(x) = 0$ → Min

→ Max

→ Saddle



$\nabla f(x, y) = 0$

* $\mathbb{R}^n$ $\{x, y, z\}$ $\{a, b, c\}$

eg: $\mathbb{R}^3$: $ax + by + z = \not{d}\ /\ 0$?

Made with Goodnotes

$\mathbb{R}^n$: $a_1x_1 + \cdots + a_nx_n = d$

hyperplane

* partial derivative

$f : \mathbb{R}^2 \to \mathbb{R}$

(x,y)

$$f_x(a,b) = \lim_{h \to 0} \frac{f(\vec{a}+h, \vec{b}) - f(a,b)}{h}$$

$$f_y(a,b) = \lim_{h \to 0} \frac{f(a, b+h) - f(a,b)}{(h)}$$

Made with Goodnotes

$\mathbb{R}^2 \to R$ $\quad f(x,y) = x^2 + y^2$

(x,y)

$d \to \partial$

$[x,y]$

$$f_x(x,y) = \frac{\partial f}{\partial x} = 2x$$

del

$$f_y(x,y) = \frac{\partial f}{\partial y} = 2y$$

$$\frac{\mathbb{R}^d \to \mathbb{R}}{(x_1 \cdots x_d)}$$

$[x_1 \cdots x_d]$

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}, \cdots, \frac{\partial f}{\partial x_d} \right)$$

Made with Goodnotes

eg: $\mathbb{R}^3 \to \mathbb{R}$

$$f(x,y,z) = x^2+y^2+z^2$$

$$\nabla f(x,y,z) = [2x, 2y, 2z]$$

$$\Rightarrow \mathcal{L} = \mathcal{L}(\theta, \theta_2, \dots \theta_n)$$

$$x_1 \; x_2 \; - - - x_n$$

$$\nabla_\theta \mathcal{L} = \left(\right)$$

Made with Goodnotes

Linear Approx^{n} in Sev. vars

$$h = [h_1 \; - - - \; h_d]_d$$

$$f(x+h) \simeq f(x) + \left(\frac{\partial f}{\partial x_1} \; - - - - \right) h$$

$$\nabla f$$

$$\simeq f(x) + \nabla f \cdot h$$

Made with Goodnotes

* Directional Derivative

↳ what if we choose to go along

$$\vec{u} = [1,2,3]$$

$$D_u f(x) = \lim_{h \to 0} \frac{f(x + hu) - f(x)}{h}$$

∇f

$u \to \{\|u| = 1\}$

$u = \hat{u}$

rate of change per unit distance in dir^n u

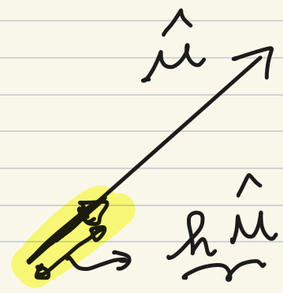
$\vec{u}(h)$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\downarrow$$

$$\nabla f = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

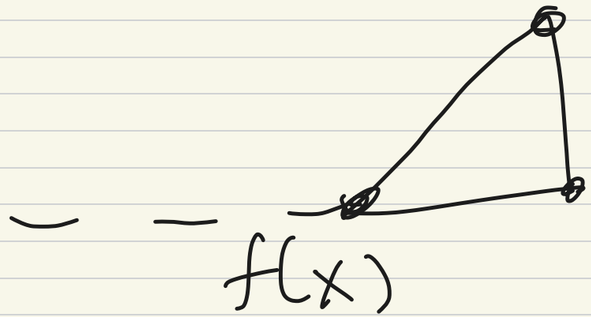
$$D_{\hat{u}}(f(x)) = \lim_{h \to 0} \frac{f(x + h \cdot \hat{u}) - f(x)}{(h)}$$

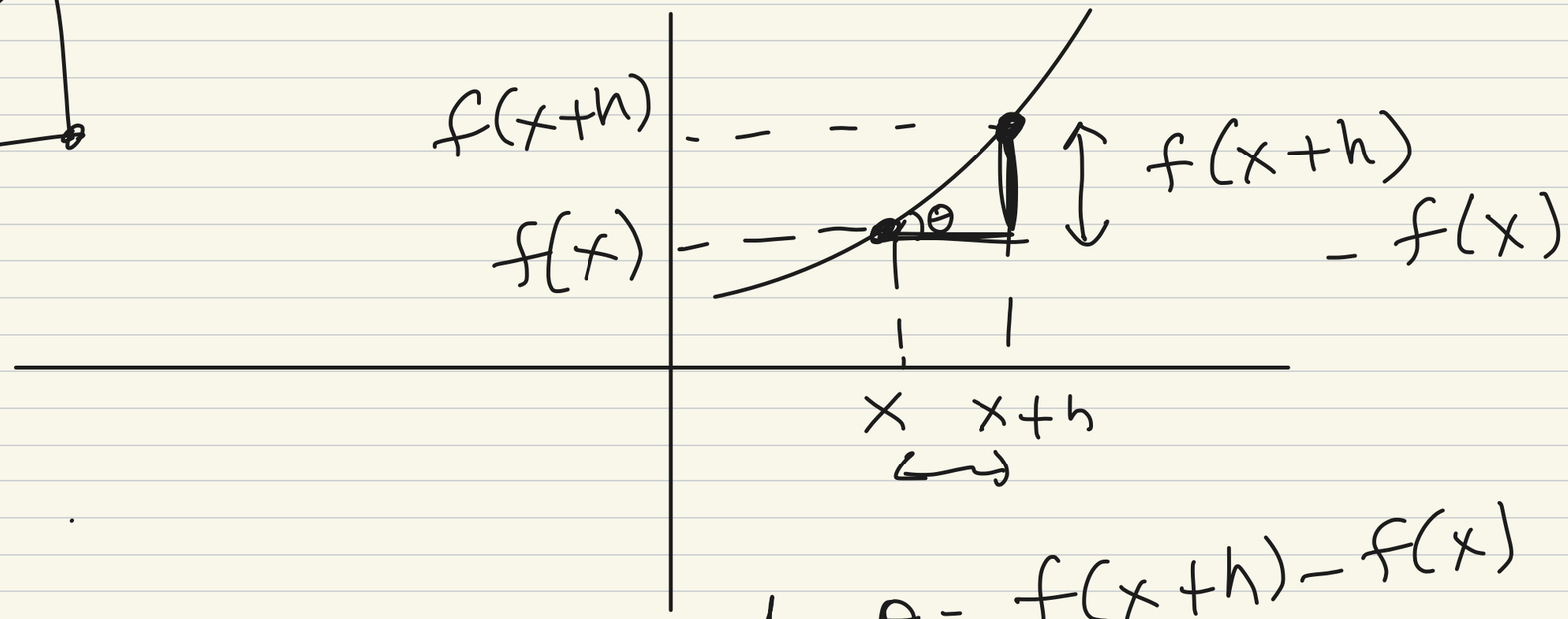


Made with Goodnotes

$$f(x+\underbrace{hu}_{r}) = \underbrace{f(x)}_{} + D_u(f(x)) \cdot h$$

$$f(x+h) = f(x) + f'(x)h$$





$$\tan\theta = \frac{f(x+h)-f(x)}{x+h-x}$$

Made with Goodnotes